Hypothesis Testing — Cheatsheet

Last-minute revision · Basics + formulas + decision rules + pitfalls

Source: M. Latif — Hypothesis Testing (Feb 2026)
89 slides → 4 pages · 6 test families

TEST	WHEN TO USE	PARAM	TEST STATISTIC	REJECT H_0 (TWO-SIDED)	DISTRIBUTION
Z-test (1-sample)	Normal pop, σ known	μ	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$\lvert Z\rvert > z_{\alpha/2}$	$N(0,1)$
t-test (1-sample)	Normal pop, σ unknown	μ	$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$\lvert T\rvert > t_{n-1,\alpha/2}$	t_{n-1}
2-sample Z	2 normals, σ^2 known	μ_x − μ_y	$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}}$	$\lvert Z\rvert > z_{\alpha/2}$	$N(0,1)$
Pooled t (2-sample)	2 normals, σ^2 unknown but equal	μ_x − μ_y	$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{1/n + 1/m}}$	$\lvert T\rvert > t_{n+m-2,\alpha/2}$	t_{n+m-2}
Welch (large n)	2 normals, σ^2 unknown & unequal, n,m large	μ_x − μ_y	$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{S_x^2/n + S_y^2/m}}$	$\lvert Z\rvert > z_{\alpha/2}$	≈ N(0,1)
Paired t	Matched pairs (before/after)	μ_w (mean diff)	$Z = \frac{\bar{W}}{S_w/\sqrt{n}}$	$\lvert Z\rvert > z_{\alpha/2}$	≈ N(0,1) (large n)
χ^2-test (variance)	Normal pop	σ^2	$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$	$\chi^2 > \chi^2_{n-1,\alpha/2}$ or $< \chi^2_{n-1,1-\alpha/2}$	χ^2_{n-1} (asymmetric)
F-test (2 variances)	2 independent normals	σ_x^2/σ_y^2	$F = \frac{S_x^2}{S_y^2}$	$F > F_{n-1,m-1,\alpha/2}$ or $< F_{n-1,m-1,1-\alpha/2}$	$F_{n-1,m-1}$ (asymmetric)
Binomial exact	Small n, Bernoulli data	p	$p\text{-val} = \sum_{i \ge x}\binom{n}{i}p_0^i(1-p_0)^{n-i}$	Reject if $p\text{-val} \le \alpha$	$B(n, p_0)$
Prop. Z-test	Large n ($np_0>5$, $n(1-p_0)>5$)	p	$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}}$	$\lvert Z\rvert > z_{\alpha/2}$	≈ N(0,1)
Fisher-Irwin	2 Bernoulli samples, exact	$p_1 = p_2$	$P(X_1 = i \mid X_1 + X_2 = k) = \frac{\binom{n_1}{i}\binom{n_2}{k-i}}{\binom{n_1+n_2}{k}}$	Reject if $P(X_1 \ge x_1) \le \alpha/2$ or $P(X_1 \le x_1) \le \alpha/2$	Hypergeometric
2-prop Z-test	2 large Bernoulli samples	$p_1 - p_2$	$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n+1/m)}}, \hat{p} = \frac{X+Y}{n+m}$	$\lvert Z\rvert > z_{\alpha/2}$	≈ N(0,1)

1 FOUNDATIONS — VOCABULARY YOU MUST OWN

slides 4-13

CONCEPT What is a hypothesis test?

A **statistical hypothesis** is a statement about a parameter θ of a population distribution. We draw a sample and ask: *is the sample consistent with the claim?* If yes → "accept" (really: fail to reject). If no → reject.

WHY "Accepting" does NOT mean proving true — it only means the data does not contradict the hypothesis. We never prove H_0; we only fail to refute it.

CONCEPT Null vs Alternative hypothesis

- **Null H_0**: the "status quo" / no-effect / no-difference claim. Usually what we want to disprove.
- **Alternative H_1**: what we suspect is true; rejecting H_0 means H_1 is supported.
- **Simple H_0**: pins a single value, e.g. $H_0 : \theta = 1$. **Composite H_0**: a range, e.g. $H_0 : \theta \ge 1$.
- Rejection of H_0 ⟺ acceptance of H_1.

CONCEPT Critical region & test statistic

A test partitions sample space into **acceptance region** and **critical (rejection) region** C. We reject H_0 iff $(X_1, \ldots, X_n) \in C$.

$$C = \{(X_1, \ldots, X_n) : |T_{\text{stat}}| > c\}$$

The constant c is the **critical value**, chosen so that P(reject | H_0 true) = α.

CONCEPT Type I vs Type II error

	H_0 TRUE	H_0 FALSE
Reject H_0	Type I (α)	Correct (1−β)
Accept H_0	Correct (1−α)	Type II (β)

α = P(Type I) — significance level. Fixed in advance (0.10, 0.05, 0.01).
β(μ) = P(Type II | μ) — depends on true μ. **Power = 1 − β(μ)**.

CONCEPT p-value — the smallest α that rejects

The **p-value** is the probability, under H_0, of observing a test statistic at least as extreme as the one observed.

$$p = P(\text{observe } |T| \ge |t_{\text{obs}}| \mid H_0 \text{ true})$$

RULE Reject H_0 at level α **iff** $p \le \alpha$. Smaller p ⟹ stronger evidence against H_0.

e.g. p = 0.093 → reject at α=0.10 but NOT at α=0.05.

CONCEPT OC curve & Power function

Operating Characteristic (OC) curve: β(μ) plotted vs true μ. = P(accept H_0 | μ).

Power function: $1 - \beta(\mu)$ = P(reject H_0 | μ). Power ↑ as true μ moves away from μ_0.

USE Power tells you how likely your test is to *detect* a real effect of size μ_1. Low power ⟹ test misses real effects.

CONCEPT Confidence interval ↔ Hypothesis test

A $100(1-\alpha)\%$ CI for μ is $\bar{X} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$. Under H_0, this CI contains μ_0 with prob $1-\alpha$.

$$\text{Reject } H_0 : \mu = \mu_0 \text{ (two-sided, level } \alpha) \iff \mu_0 \notin \left(\bar{X} - z_{\alpha/2}\tfrac{\sigma}{\sqrt{n}},\ \bar{X} + z_{\alpha/2}\tfrac{\sigma}{\sqrt{n}}\right)$$

PITFALL This equivalence only holds for **two-sided** tests. One-sided tests need one-sided CIs.

CONCEPT Decision tree — which test?

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Q1: How many populations?
  1 → Q2
  2 → Q5

Q2: What parameter?
  μ   → Q3
  σ² → χ²-test (1-pop)
  p   → Q4

Q3: σ known?
  Y → Z-test
  N → t-test

Q4: n large?
  Y → Z-test for p
  N → exact binomial

Q5: What parameter?
  μ_x-μ_y → Q6
  σ_x²=σ_y² → F-test
  p_x=p_y → Q7

Q6: Paired or independent?
  paired → paired t-test
  indep → Q6b
  Q6b: σ² known?
    Y → 2-sample Z
    N → equal var? Y→pooled t N→Welch (large n)

Q7: Both n,m large?
  Y → 2-prop Z
  N → Fisher-Irwin
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2 ONE-SAMPLE MEAN TESTS — Z & T

slides 14-44

Z Z-test — σ² known

ASSUME $X_1,\dots,X_n \sim N(\mu,\sigma^2)$, σ known, μ_0 fixed.

STAT

$$Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \text{ under } H_0$$

H_1	Reject H_0 if	p-value
$\mu \neq \mu_0$	$\lvert Z\rvert > z_{\alpha/2}$	$2P(Z \geq \lvert z\rvert)$
$\mu > \mu_0$	$Z > z_\alpha$	$P(Z \geq z)$
$\mu < \mu_0$	$Z < -z_\alpha$	$P(Z \leq z)$

WHY $\bar{X} \sim N(\mu,\sigma^2/n)$. Under H_0 standardizing gives N(0,1).

EX Signal: σ=2, n=5, $\bar{X}$=9.5, $H_0: \mu = 8$. $Z = (9.5-8)/(2/\sqrt{5}) = 1.68$. At α=0.05, $z_{0.025} = 1.96$, |Z|<1.96 → **fail to reject**. At α=0.10, $z_{0.05} = 1.64$ → **reject**. p=0.093.

Z Critical values to memorize

α	z_α (1-side)	z_{α/2} (2-side)
0.10	1.282	1.645
0.05	1.645	1.960
0.025	1.960	2.241
0.01	2.326	2.576

Common z-scores you must NOT look up in exam: 1.645, 1.96, 2.576.

t t-test — σ² unknown

ASSUME $X_1,\dots,X_n \sim N(\mu,\sigma^2)$, both μ & σ² unknown. Estimate σ² by $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$.

STAT

$$T = \frac{\bar{X}-\mu_0}{S/\sqrt{n}} \sim t_{n-1} \text{ under } H_0$$

H_1	Reject H_0 if	p-value
$\mu \neq \mu_0$	$\lvert T\rvert > t_{n-1,\alpha/2}$	$2P(T_{n-1} \geq \lvert t\rvert)$
$\mu > \mu_0$	$T > t_{n-1,\alpha}$	$P(T_{n-1} \geq t)$
$\mu < \mu_0$	$T < -t_{n-1,\alpha}$	$P(T_{n-1} \leq t)$

WHY Replacing σ by S adds uncertainty → distribution gets heavier tails than N(0,1). df = n−1 because S^2 uses one degree of freedom to estimate $\bar{X}$. As n→∞, $t_{n-1} \to N(0,1)$.

EX Cigarettes: n=20, $\bar{X}$=1.54, S=0.8, $H_0: \mu = 1.6$ vs $H_1: \mu < 1.6$, α=0.05. $T = (1.54-1.6)/(0.8/\sqrt{20}) = -0.335$. $-t_{19,0.05} \approx -1.729$. T > −1.729 → **fail to reject**.

PITFALL Z vs t — common exam traps

- **σ unknown → MUST use t, not Z** (even if n is large, t is safer; Z is only approximate).
- **df for t = n − 1** for 1-sample, **n + m − 2** for pooled 2-sample.
- **t-table is symmetric**: $t_{\nu,1-\alpha} = -t_{\nu,\alpha}$. χ^2 and F are NOT.
- "Accept H_0" really means "fail to reject" — never write "we proved H_0 is true".
- Two-sided p-value is **2 ×** one-sided; one-sided p = $P(Z \geq z)$ or $P(Z \leq z)$.
- p-value is the **smallest α** at which you can reject, NOT the probability that H_0 is true.

3 TWO-SAMPLE TESTS & PAIRED DATA

slides 45-58

2-SAMPLE Case I — variances known

ASSUME $X_i \sim N(\mu_x,\sigma_x^2)$, $Y_j \sim N(\mu_y,\sigma_y^2)$, σ² known, independent samples of size n & m.

$$Z = \frac{\bar{X}-\bar{Y}}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}} \sim N(0,1) \text{ under } H_0$$

REJECT $\lvert Z\rvert > z_{\alpha/2}$ (two-sided) · p = $2P(Z \geq \lvert z\rvert)$.

EX 8.4A Tyres A (n=10, $\bar{X}$=61.65, σ_x=4) vs B (m=8, $\bar{Y}$=60.03, σ_y=6). $Z = 1.52/(0.446) = 0.679$. $\lvert Z\rvert = 0.679 < 1.96$ → fail to reject H_0: μ_A = μ_B at 5%.

2-SAMPLE Case II — variances unknown, assumed equal (pooled t)

ASSUME σ_x² = σ_y² = σ² (unknown). Pool both samples to estimate σ².

$$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}, \quad T = \frac{\bar{X}-\bar{Y}}{S_p\sqrt{1/n + 1/m}} \sim t_{n+m-2}$$

REJECT $\lvert T\rvert > t_{n+m-2,\alpha/2}$. df = **n+m−2**. p = $2P(T_{n+m-2} \geq \lvert t\rvert)$.

WHY Pooling gives a better variance estimate, so we get more degrees of freedom than running two t-tests separately.

2-SAMPLE Case III — variances unknown, unequal (Welch, large n)

ASSUME σ_x² ≠ σ_y², both unknown, n & m **large**.

$$Z = \frac{\bar{X}-\bar{Y}}{\sqrt{S_x^2/n + S_y^2/m}} \approx N(0,1) \text{ under } H_0$$

REJECT $\lvert Z\rvert > z_{\alpha/2}$.

PITFALL Only valid for large n, m. For small samples with unequal variances use Welch–Satterthwaite df (not in this PDF).

PAIRED Paired t-test — before/after, matched pairs

ASSUME Pairs (X_i, Y_i); define $W_i = X_i - Y_i$ (or $Y - X$). Assume $W_i \sim N(\mu_w,\sigma_w^2)$.

STAT (treat W as a 1-sample problem!)

$$Z = \frac{\bar{W}}{S_w/\sqrt{n}}, \quad \bar{W} = \tfrac{1}{n}\sum W_i, \quad S_w^2 = \tfrac{1}{n-1}\sum(W_i - \bar{W})^2$$

REJECT $\lvert Z\rvert > z_{\alpha/2}$. p = $2P(Z \geq \lvert z\rvert)$.

WHY Pairing removes between-subject variance. Each subject is its own control → tighter test.

EX 8.4D Safety program: $\bar{W} = -2.15$, S_w=3, n=10. $Z = -2.15/(3/\sqrt{10}) = -2.266$. $\lvert Z\rvert = 2.266 > 1.96$ → **reject H_0** at 5%. Program has significant effect.

PITFALL Paired vs independent — pick wrong, get wrong answer

- **Paired**: same subject measured twice (before/after, left/right). Use differences W_i.
- **Independent**: two separate groups. Use 2-sample Z / pooled t / Welch.
- Never run an independent-samples test on paired data — you lose power.
- Sign of W depends on whether you compute X−Y or Y−X — be consistent with H_1 direction.

4 VARIANCE TESTS — X² & F

slides 59-67

χ² One-population variance test

ASSUME $X_1,\dots,X_n \sim N(\mu,\sigma^2)$. Test $H_0: \sigma^2 = \sigma_0^2$.

$$\chi^2_{\text{stat}} = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1} \text{ under } H_0$$

REJECT (two-sided): $\chi^2_{\text{stat}} > \chi^2_{n-1,\alpha/2}$ **OR** $\chi^2_{\text{stat}} < \chi^2_{n-1,1-\alpha/2}$.

p = $2\min\{P(\chi^2_{n-1} < c),\ 1 - P(\chi^2_{n-1} < c)\}$.

KEY χ^2 is **NOT symmetric**! Lower & upper critical values differ in magnitude. Don't write $\pm c$.

F Equality of two variances

ASSUME $X_i \sim N(\mu_x,\sigma_x^2)$, $Y_j \sim N(\mu_y,\sigma_y^2)$, independent. Test $H_0: \sigma_x^2 = \sigma_y^2$.

$$F = \frac{S_x^2/\sigma_x^2}{S_y^2/\sigma_y^2} = \frac{S_x^2}{S_y^2} \sim F_{n-1,\,m-1} \text{ under } H_0$$

REJECT $F > F_{n-1,m-1,\alpha/2}$ **OR** $F < F_{n-1,m-1,1-\alpha/2}$.

p = $2\min\{P(F_{n-1,m-1} < F),\ 1 - P(F_{n-1,m-1} < F)\}$.

WHY If σ_x²=σ_y², then S_x^2/S_y^2 should be ≈ 1. The F-distribution is the ratio of two independent χ^2/df.

PITFALL F is **asymmetric** & only positive. Convention: put larger S^2 in numerator so F ⩾ 1, then use upper tail only.

EX 8.5B Catalyst 1: n=10, $S_1^2 = 0.14$. Catalyst 2: m=12, $S_2^2 = 0.28$. $F = 0.28/0.14 = 2$. Critical $F_{9,11,0.025} \approx 3.59$. 2 < 3.59 → **fail to reject** H_0 at 5%.

5 BERNOULLI / PROPORTION TESTS

slides 68-82

BIN **One proportion — exact binomial test**

ASSUME $X \sim B(n,p)$ = # of "successes" (defects) in n independent Bernoulli trials. Test $H_0 : p \le p_0$ vs $H_1 : p > p_0$.

STAT Under $H_0 : p = p_0$, $X \sim B(n, p_0)$.

REJECT H_0 if p-value $\le \alpha$, where:

$$p\text{-val} = P(X \ge x \mid p = p_0) = \sum_{i=x}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i}$$

USE When n is small or p_0 near 0/1 — normal approximation unreliable.

EX 8.6A Chips: n=300, x=10 defectives, $H_0 : p \le 0.02$. p-val = $\sum_{i=10}^{300} \binom{300}{i} (0.02)^i (0.98)^{300-i} = 0.0818$. p > 0.05 → **fail to reject**.

BIN **One proportion — approximate Z-test**

ASSUME n **large**: rule of thumb $np_0 > 5$ AND $n(1-p_0) > 5$. Then $X \approx N(np_0, np_0(1-p_0))$.

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} \approx N(0,1) \text{ under } H_0$$

H_1	Reject if	p-value
$p \neq p_0$	$\lvert Z \rvert > z_{\alpha/2}$	$2P(Z \ge \lvert z \rvert)$
$p > p_0$	$Z > z_\alpha$	$P(Z \ge z)$
$p < p_0$	$Z < -z_\alpha$	$P(Z \le z)$

EX 8.6A Same data: $np_0 = 6 > 5$ ✓. $Z = (10-6)/\sqrt{5.88} = 1.443$. p = $P(Z > 1.44) = 0.0749$. Compare to exact 0.0818 — close but not identical.

BIN **Two proportions — Fisher-Irwin (exact)**

ASSUME $X_1 \sim B(n_1, p_1)$, $X_2 \sim B(n_2, p_2)$, independent. Test $H_0 : p_1 = p_2 = p$.

Under H_0, conditional on $X_1 + X_2 = k$, the distribution of X_1 is **hypergeometric**:

$$P(X_1 = i \mid X_1 + X_2 = k) = \frac{\binom{n_1}{i}\binom{n_2}{k-i}}{\binom{n_1+n_2}{k}}$$

REJECT H_0 (two-sided, level α) if $P(X_1 \ge x_1 \mid k) \le \alpha/2$ **or** $P(X_1 \le x_1 \mid k) \le \alpha/2$.

USE Small samples where normal approx fails. **Fisher's exact test** in software = this.

BIN **Two proportions — approximate Z-test**

ASSUME n, m both large. $\hat{p}_1 = X/n$, $\hat{p}_2 = Y/m$, pooled $\hat{p} = (X+Y)/(n+m)$.

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n} + \frac{1}{m}\right)}} \approx N(0,1) \text{ under } H_0$$

REJECT $\lvert Z \rvert > z_{\alpha/2}$. p = $2P(Z \ge \lvert z \rvert)$.

WHY POOL? Under H_0 both samples come from same p, so we estimate the common p once (more data → tighter SE) rather than using two separate $\hat{p}_1, \hat{p}_2$.

6 EXAM SURVIVAL KIT — WORKFLOW & UNIVERSAL PITFALLS

rapid recall

WORKFLOW **Universal 6-step recipe for any hypothesis test**

1. **State H_0 & H_1** — write as equations, not words. H_0 always has "=".
2. **Identify parameter & distribution** — μ? σ²? p? → Z / t / χ^2 / F / B(n,p).
3. **Check assumptions** — normal? σ known? n large? independent? paired?
4. **Compute test statistic** — plug in numbers carefully. Show units.
5. **Compare to critical value** OR compute p-value (NOT both — pick one).
6. **State conclusion in plain English** — "reject/fail to reject H_0 at α level", then interpret.

PITFALLS **10 universal traps**

- **1.** σ unknown → use t, NOT Z (even when n is large).
- **2.** Two-sided p = 2 × one-sided p. Forgetting the "2×" is the #1 p-value mistake.
- **3.** χ^2 and F are **not symmetric** — never write ±c. Use both upper & lower critical values.
- **4.** df: 1-sample t → n−1 · pooled 2-sample t → n+m−2 · χ^2 → n−1 · F → (n−1, m−1).
- **5.** "Fail to reject" ≠ "accept H_0 is true". H_0 is never proven; only not-refuted.
- **6.** p-value is NOT P(H_0 true | data). It is P(data | H_0 true).
- **7.** Paired data → use differences W_i, NOT 2-sample test.
- **8.** One-sided H_1 → use z_α (NOT z_{α/2}); reject in only ONE tail.
- **9.** $np_0 > 5$ and $n(1-p_0) > 5$ required before using normal approx to binomial.
- **10.** For F-test, put LARGER sample variance in numerator so $F \geqslant 1$ → only check upper critical value.

CONCEPT **Significance vs Effect size vs Power**

- **α (significance)**: how willing you are to falsely reject (false positive rate).
- **Effect size**: how different μ_1 is from μ_0 in real units — drives β.
- **Power = 1−β**: probability you'll detect a true effect. Increase by ↑n, ↑α, or ↑$\lvert\mu_1 - \mu_0\rvert$.
- **Small p, large n** ≠ big effect. With huge n even tiny effects become "significant".

WHY A test can be statistically significant yet practically meaningless. Always pair p-value with effect size & CI.

CONCEPT **One-sided vs two-sided — when to use which**

H_1 shape	Reject if	Use when
$\mu \neq \mu_0$ (2-sided)	$\lvert Z \rvert > z_{\alpha/2}$	Default; both directions surprise you
$\mu > \mu_0$ (1-sided)	$Z > z_\alpha$	Pre-registered: only "more than" is interesting
$\mu < \mu_0$ (1-sided)	$Z < -z_\alpha$	Pre-registered: only "less than" is interesting

PITFALL Don't switch to one-sided after seeing the data — that inflates Type I error. Decide **before** looking at results.

CONCEPT **Connection: CI ↔ Test (only 2-sided)**

A $100(1-\alpha)\%$ two-sided CI for μ contains μ_0 **iff** the two-sided test at level α fails to reject H_0: μ = μ_0.

$\bar{X} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$ — if μ_0 inside → fail to reject; if outside → reject.

PITFALL CI ↔ test link breaks for one-sided tests. One-sided tests need one-sided (upper or lower bound) CIs.

CONCEPT **Reading a t-table (mental model)**

- Rows = degrees of freedom (df). Columns = upper-tail probability α.
- $t_{\nu,\alpha}$ = value with area α to the **right**.
- Symmetry: $t_{\nu,1-\alpha} = -t_{\nu,\alpha}$. So $-t_{\nu,0.95} = t_{\nu,0.05}$.
- As df → ∞, $t_{\nu,\alpha} \to z_\alpha$. Memorize: $t_{\infty,0.025} = 1.96 = z_{0.025}$.
- For df not in table: use the **next smaller** df (conservative — gives larger critical value, harder to reject).

FINAL **If you forget everything else, remember this**

1. Pick test by: ① parameter type ② # of populations ③ σ known? ④ paired? ⑤ n large?
2. Statistic = (estimate − null)/(SE of estimate).
3. Reject H_0 if |stat| > critical OR p ⩽ α. **Two-sided → multiply one-tail p by 2.**
4. σ unknown → t. Variance test → χ^2 or F (asymmetric!). Small n, p-test → exact binomial.
5. Always answer in plain English: "We [reject/fail to reject] H_0 at the α level. There [is/is not] sufficient evidence that ..."